

Analysis of Elite and Public Discourse on Data Center Construction in Pennsylvania

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Motivation & Research Question

The Problem

- ▶ AI infrastructure expanding rapidly across the U.S.
- ▶ Data centers demand extraordinary electricity, water, and land
- ▶ Policymakers respond with selective discursive strategies

Why Pennsylvania?

- ▶ Most active data center market on the East Coast
- ▶ Contested energy landscape: natural gas + renewables
- ▶ Perennial **swing state** → pluralistic discursive environment

Research Question

How do Pennsylvania state officials **frame** data center construction in official communications?

Key Argument

Elite framing is **strategic** — selectively addressing certain concerns while maintaining deliberate silence on others, calibrated to political context.

Data Collection: pa.gov via Coveo API

Source & Method

- ▶ Commonwealth of Pennsylvania public portal (pa.gov)
- ▶ Internal search API (**Coveo**) identified via browser network inspection
- ▶ HTTP POST requests replicated in Python
- ▶ Full text extracted via BeautifulSoup

Three Search Queries

"data center" "power plant" "AI infrastructure"

Agencies: Governor's Office, DEP, DCED

Processing Steps

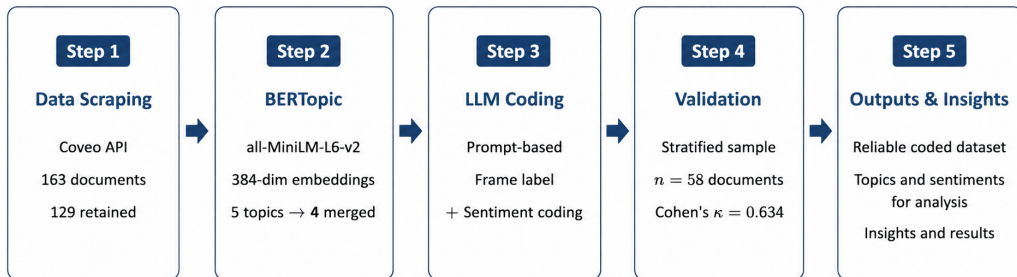
1. Retrieve structured JSON results
2. Remove duplicates by URL
3. Relevance screening via LLM (Claude)
4. Extract: title, URL, year, agency, full text
5. Light preprocessing

Final Corpus

163 documents collected (2023–2026)

129 documents retained after screening

Pipeline: Elite Rhetoric



Coding Validation: LLM vs. Human

Agreement Statistics

- ▶ Overall accuracy: **72%** (42/58 documents)
- ▶ Cohen's $\kappa = 0.634$ — substantial agreement
- ▶ Exceeds conventional threshold of 0.60

Performance by Frame

- ▶ **Electricity**: 11/12 correct (*strongest*)
- ▶ **Economic**: 12/13 correct
- ▶ **Irrelevant**: 14/26 correct (*weakest*)

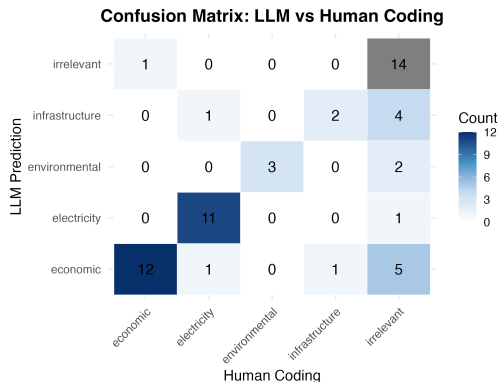
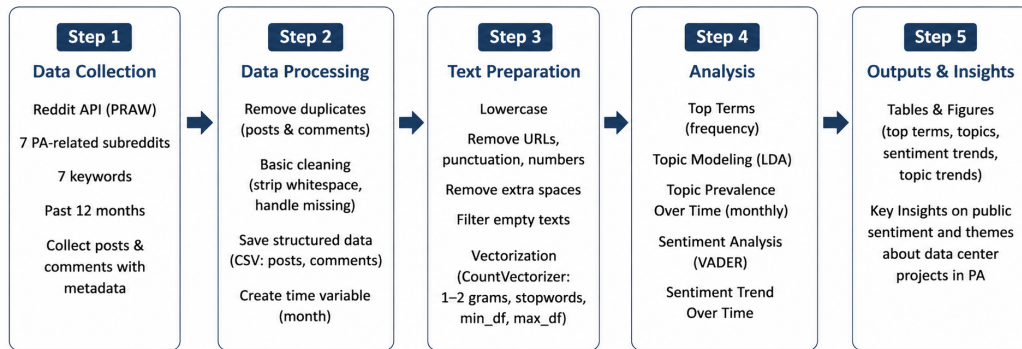


Figure: Confusion Matrix: LLM vs. Human Coding

Pipeline: Public Discourse



Finding 1: Economic Frame Dominates

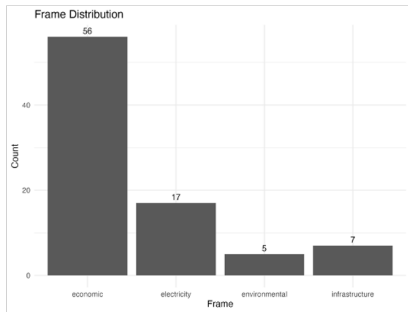


Figure: Overall Frame Distribution

Distribution

- ▶ **Economic** (n=56): dominant frame
- ▶ **Electricity** (n=17): second most prominent
- ▶ **Infrastructure** (n=7), **Environmental** (n=5): marginal

Key Phrases — Economic Frame

- ▶ *jobs, investment, tax revenue*
- ▶ *largest private sector investment in Pennsylvania history*
- ▶ *1,250 new high-paying high-tech jobs*

→ Data centers consistently framed as **economic opportunity**, not environmental risk

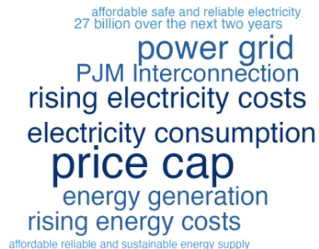
Finding 1: Economic Frame Dominates (cont.)



A word cloud for the economic topic. The words are arranged in a vertical stack, with 'jobs' being the largest and most prominent. Other words include 'investment', 'tax revenue', 'economic development', 'GDP growth', '12 billion in private sector investments', '\$529 million investment', 'largest private sector investment in Pennsylvania history', '\$22 million in the project through grants and capital funding', and '1250 new high-paying high-tech jobs'.

1250 new high-paying high-tech jobs
\$22 million in the project through grants and capital funding
largest private sector investment in Pennsylvania history
\$529 million investment
jobs
investment
tax revenue
economic development
GDP growth
12 billion in private sector investments

Top terms in economic topic.



A word cloud for the electricity cost topic. The words are arranged in a vertical stack, with 'price cap' being the largest and most prominent. Other words include 'energy generation', 'rising energy costs', 'electricity consumption', 'rising electricity costs', 'PJM Interconnection', 'power grid', '27 billion over the next two years', 'affordable safe and reliable electricity', and 'affordable reliable and sustainable energy supply'.

affordable safe and reliable electricity
27 billion over the next two years
power grid
PJM Interconnection
rising electricity costs
electricity consumption
price cap
energy generation
rising energy costs
affordable reliable and sustainable energy supply

Topic terms in electricity cost topic.

Finding 2: Strategic Consumer Protection Narrative

The Electricity Frame constructs a three-actor populist structure:

VILLAIN

**PJM Interconnection
& Tech Companies**

“opaque, self-serving rules”
“corporate cost-shifting”

VICTIM

**Ordinary Pennsylvania
Households**

“67 million consumers”
“\$27 billion in costs”

HERO

Governor Shapiro

Price cap, lawsuit,
PJM reform

Key Rhetorical Move

Data centers are **welcomed** (economic growth) but must “pay their own way” — reconciling pro-investment and pro-consumer positions within a single **populist narrative**.

Finding 3: Framing Shifts within Electoral Cycle

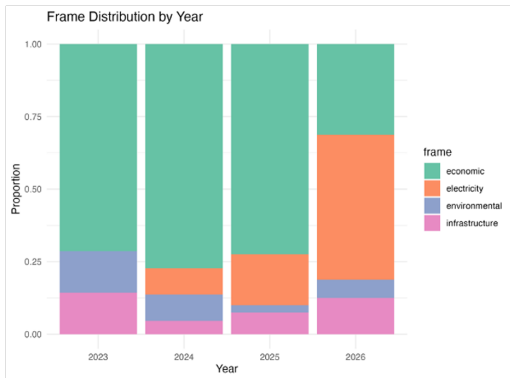


Figure: Frame Distribution by Year (2023–2026)

Temporal Pattern

- ▶ 2023–2025: Economic topic >75% each year (2024 presidential election)
- ▶ 2026 (mid-term elections): Electricity topic rises sharply to ~40%

Implications

- ▶ 2023–2025: attract investment → economic frame serves recruitment
- ▶ 2026: energy affordability becomes **electorally salient**
- ▶ Framing tracks **political incentives**, not merely policy events

→ Consistent with strategic communication theory: political actors adjust issue emphasis during **heightened electoral accountability**

Conclusion

Main Findings

1. **Economic frame dominates:** data center development framed as jobs and investment
2. **Strategic silence:** environmental and community costs absent from official discourse
3. **Populist consumer narrative:** PJM as villain; Shapiro as protector
4. **Electoral calibration:** frame distribution shifts in 2026 election year

Methodological Contribution

BERTopic + LLM coding enables systematic framing analysis at scale

Cohen's $\kappa = 0.634$ confirms reliability

Broader Implication

Elite framing shapes public understanding of AI infrastructure trade-offs — rendering costs invisible while amplifying benefits